

Dataset Description and Statistical Overview

Thyroid Disease Classification Dataset - Bangladesh Clinical Case Study

1. Dataset Access and Availability

The thyroid dataset used in this study was collected from patients who visited the National Institute of Nuclear Medicine and Allied Sciences (NINMAS), Dhaka, Bangladesh, for thyroid testing between October and December 2024. The dataset is not publicly available because of privacy and institutional restrictions. According to the manuscript, the data may be requested from the corresponding author on reasonable request, subject to institutional permission and privacy requirements.

2. Ethical and Data Collection Summary

Item	Description
Ethics approval	Approved by the IRB committee of the Department of Electrical and Electronic Engineering, CUET.
Institutional permission	Data collection permission was approved by NINMAS, Dhaka, Bangladesh.
Consent	Informed consent was obtained from all individual participants included in the study.
Dataset type	Single-center real clinical patient-record dataset.
Availability	Not publicly available; available from the corresponding author on reasonable request.

3. Dataset Composition

Attribute	Value / Description
Total patient records	348
Target classes	5 classes: euthyroid, clinical hyperthyroid, clinical hypothyroid, subclinical hyperthyroid, subclinical hypothyroid.
Original attributes	29 original attributes including diagnosis-related fields; predictive modeling used curated features after leakage-prone fields were removed.
Feature categories	Demographics, medication history, symptoms, physical conditions, hormonal markers, and blood-test results.
Key numerical markers	Age, TSH Value, FT3 Value, FT4 Value.
Important clinical predictors	TSH, FT4, FT3, bowel condition, weight change, skin problem, tremor, sleep disorder, memory problem, medication-related variables.

4. Class Distribution

Class	Records	Percentage
Euthyroid (EU)	217	62.36%
Clinical hyperthyroid (HYPER)	51	14.66%
Subclinical hyperthyroid (SUB-HYPER)	33	9.48%
Clinical hypothyroid (HYPO)	28	8.05%
Subclinical hypothyroid (SUB-HYPO)	19	5.46%
Total	348	100%

Observation: The dataset is imbalanced, with euthyroid cases forming the dominant class and subclinical hypothyroid cases forming the smallest class. This imbalance motivated training-fold-only SMOTE during model development.

5. Numerical Attribute Summary

Statistic	Age	TSH Value	FT3 Value	FT4 Value
Count	348	348	183	298
Mean	40.16	5.30	5.71	16.72
Std	16.20	16.59	3.52	6.04
Min	9	0.01	1.25	2.69
25%	27	0.39	4.10	13.76
Median	38	1.47	4.73	15.82
75%	49	3.01	5.90	19.47
Max	85	150	30	73.53

6. Missing-Value Overview

Variable	Available values	Missing values	Missing rate
Age	348	0	0.0%
TSH Value	348	0	0.0%
FT3 Value	183	165	47.4%
FT4 Value	298	50	14.3%

KNN imputation with $k = 4$ was used for numerical missing values because it preserves local data structure better than simple mean, median, or mode imputation in this clinical setting.

7. Leakage-Safe Train-Test and Validation Design

Component	Description
Train-test split	70:30 patient-disjoint stratified split.
Training set	244 patient records.
Independent test set	104 patient records.
Cross-validation	10-fold stratified cross-validation on the training set only.
Feature selection	VT-MI applied inside the cross-validation/training pipeline; top 15 features retained.
Class balancing	SMOTE applied only to training folds, not to validation or test data.
Final evaluation	Final model retrained on the full training set and evaluated once on the isolated hold-out test set.

8. Dataset-Level Modeling Outcome Summary

Pipeline	Best model	Accuracy	Macro-F1	Weighted-F1	Kappa	MCC
Full features	CatBoost / Bagging	96.15%	94.90%	96.09%	93.29%	93.37%
VT-MI selected features	Bagging	96.15%	94.90%	96.09%	93.29%	93.37%
VT-MI selected features	Random Forest	96.15%	92.83%	96.13%	93.35%	93.43%

9. Interpretation and Caution

- The dataset is clinically valuable because it represents local Bangladesh patient records rather than a public benchmark-only setting.
- The sample size is limited and single-center; therefore, the framework should be treated as internally validated and preliminary.
- External validation using larger, multi-center clinical cohorts is necessary before any practical clinical deployment.
- SHAP and LIME explanations identified TSH and FT4 as dominant model drivers, consistent with thyroid pathophysiology, but explanations should support - not replace - clinical judgment.

Source note: This compact statistical overview was prepared from the submitted thyroid disease classification manuscript and its dataset, preprocessing, EDA, and results sections.